Project Documentation

Adaptive AI Personal Life & Study Coach

An Agentic and Generative AI system that learns how each student works, predicts where they will struggle, and adapts its strategy.

Course: AI Agentic and Generative AI
Project track: Option C — Advanced Graduation Project
Document type: Project Documentation
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1. Executive Summary

Most study planners and AI assistants produce a plan once and then reschedule when the user falls behind. They treat every user the same and react only after a failure has already happened.

This project proposes a **Generative AI Personal Life & Study Coach** built around one central innovation: an **Adaptive Personal Model**. The system builds and continuously refines a model of how each individual user studies and achieves goals. It then uses that model to predict problems before they happen and to change its *strategy*, not just the schedule.

> **Unique Selling Proposition:** An AI system that doesn't just create a plan for you, but continuously learns how you work, predicts where you might struggle, and adapts its strategy to help you achieve your goals.

The system is composed of five cooperating components: the Adaptive Manager Agent, the Assessment Agent, the Goal & Life Coach Agent, the Personal Planning Agent, and a Generative LLM layer that powers planning, tutoring, assessment and personalization.

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2. Problem Statement

Students face several recurring problems:

1. **Generic planning.** Planners create "best in general" schedules that ignore individual differences in focus, energy, and habits.
2. **Reactive behavior.** Existing tools reschedule after a task is missed. They do not anticipate failure.

3. **Inaccurate time estimation.** Users consistently misjudge how long tasks take, so plans are unrealistic from day one.
4. **Disconnected learning and planning.** Quiz results and study progress rarely influence the schedule or the learning strategy.
5. **Hidden weaknesses.** Students often do not know which subjects, topics, or task types they are weakest at.
6. **Low consistency.** Without accountability, even good plans are abandoned.

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3. Objectives

3.1 Main Objective

Design and build an agentic, LLM-powered coaching platform whose planning and learning strategy adapt over time to each individual user through a continuously updated Personal Model.

3.2 Specific Objectives

- Convert the user's courses, exams, assignments, and goals into daily and weekly plans.
- Collect behavioral signals from actual execution (completion, timing, deferral, focus windows).
- Build and maintain a Personal Adaptive Model per user.
- Predict likely struggles (missed tasks, weak topics, overload days) before they occur.
- Adapt the strategy itself (task splitting, session timing, learning method), not only the calendar.
- Provide an AI tutor that generates quizzes, exams, and flashcards and analyzes mistakes.
- Feed assessment results back into the Personal Model and the next plan.
- Support consistency through streaks, achievements, and accountability.
- Evaluate the Adaptive Model objectively and produce improvement reports (Option C).

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4. Scope

4.1 In Scope

- Web application (HTML5, CSS3, vanilla JavaScript) with Firebase Authentication and Firestore.

- Onboarding: education details, courses per semester, exam and assignment schedules, goals.
- Five-component agent architecture with a shared Personal Model.
- Behavior tracking, struggle prediction, strategy adaptation.
- AI-generated quizzes, exams, flashcards, and error analysis.
- Gamification as a supporting layer.
- System tests, Adaptive Model evaluation, and improvement reports.

4.2 Out of Scope

- Native mobile applications.
- Integration with university information systems.
- Medical or clinical mental-health advice.
- Social or multi-user collaboration features.

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5. Innovation and Differentiation

Aspect	Typical AI Planner / Study Assistant	This Project
Plan creation	One best plan for everyone	Plan tuned to one specific person
Reaction to failure	Reschedules after missing	Predicts and prevents before missing
Learning from user	Little or none	Continuous Personal Model
Quiz and test results	Shown as scores	Feed back into planning and strategy
Gamification	Often the main feature	Supporting feature for consistency
Evaluation	Rarely measured	Measured with metrics and reports

5.1 Illustrative Example

If the system observes that the user usually fails to complete large tasks on days with many lectures, it does not wait for the failure. It proactively splits the task into smaller sessions in advance and places them in the user's most productive time windows.

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6. Core Loop

The heart of the system is a closed learning loop:

Collect Behavior → Understand User → Predict Problems → Adapt Strategy → Execute → Measure → Learn Again

Stage	Description	Main Component
Collect Behavior	Record completions, timing, deferrals, focus, quiz results	Adaptive Manager
Understand User	Update the Personal Model	Adaptive Manager
Predict Problems	Estimate risk of missing tasks or failing topics	Adaptive Manager + Assessment
Adapt Strategy	Choose how to plan and teach this user	Adaptive Manager
Execute	Generate plan, sessions, quizzes, coaching	Planning, Assessment, Coach
Measure	Compare predictions and outcomes	Adaptive Manager
Learn Again	Refine the model with new evidence	Adaptive Manager

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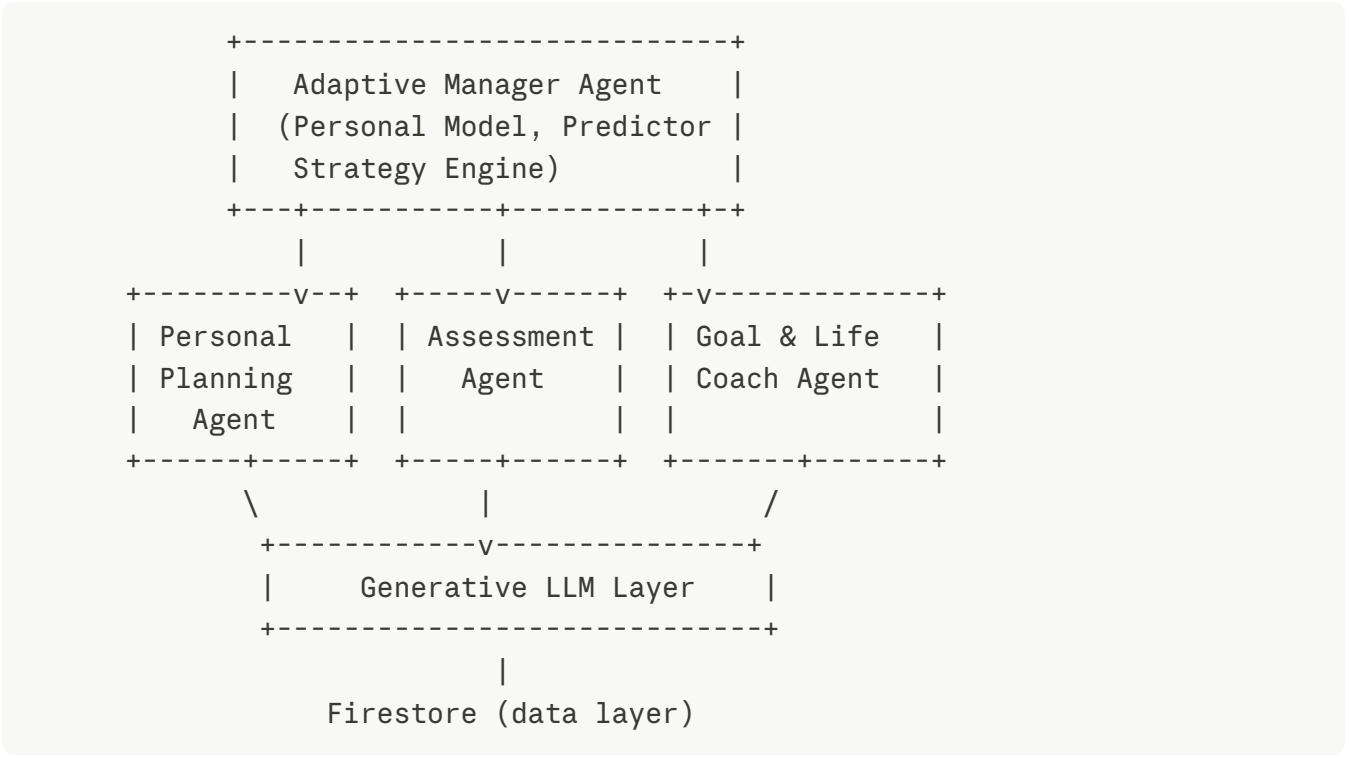
7. System Architecture

7.1 Layered Overview

1. **Presentation layer** — Web UI (vanilla HTML/CSS/JS): onboarding, dashboard, planner, study room, quizzes, progress and reports.
2. **Agent layer** — Five cooperating components coordinated by the Adaptive Manager.
3. **Intelligence layer** — Generative LLM accessed through a secure backend endpoint; prompt templates; structured JSON outputs.
4. **Data layer** — Firebase Authentication and Cloud Firestore holding the user profile, behavior events, Personal Model, plans, assessments, and reports.

7.2 Agent Coordination

The **Adaptive Manager Agent** acts as the orchestrator and owner of the Personal Model. Other agents read the model, perform their task, and report outcomes back to the Manager. No agent changes the Personal Model directly except through the Manager.



7.3 Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, vanilla JavaScript, Lucide icons, GSAP for animation
Authentication	Firebase Authentication
Database	Cloud Firestore (all data, including profile image stored as a compressed string; no Firebase Storage)
Generative AI	LLM API called through a secure backend function or proxy
Analytics and evaluation	Python scripts / notebooks for offline evaluation and report generation
Version control	GitHub

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8. The Personal Adaptive Model

The Personal Adaptive Model is a structured, continuously updated profile of how the user works. It is the project's main innovation.

8.1 Model Dimensions

Dimension	What is learned	Signals used
Productivity windows	When the user is most productive	Completion rate and session quality per hour/day
Focus capacity	How long the user can focus per session	Session length before breaks, early stops
Deferral patterns	Which task types get postponed	Reschedule counts by task type and course
Estimation bias	Gap between planned and actual duration	Estimated vs. actual time per task type
Subject weakness	Weak courses and topics	Quiz and exam error rates, repeated mistakes
Plan adherence	Overall consistency and reasons for deviation	Completion rate, self-reported reasons
Load sensitivity	Behavior on heavy-lecture or high-load days	Completion vs. daily workload
Learning preference	Which study methods work best	Score improvement by method (quiz, flashcards, explanation)

8.2 Model Representation

The model is stored as a document in Firestore with numeric features, confidence values, and timestamps. Each feature carries a **confidence score** that grows with evidence. Low-confidence features are used cautiously and the system falls back to sensible defaults.

8.3 Update Strategy

- **Event-driven updates:** after each completed, missed, or deferred task and each assessment.
- **Exponential moving averages** give recent behavior more weight than old behavior.
- **Cold start:** initial values come from onboarding answers and population defaults, then are replaced by real behavior as data accumulates.

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9. Agent Specifications

9.1 Adaptive Manager Agent

Role: Orchestrator and owner of the Personal Model; the component that makes the system adaptive.

Responsibilities

- Collect and validate behavior events from the other agents and the UI.
- Update the Personal Model.
- Run the Struggle Predictor and issue risk alerts.
- Select and apply adaptation strategies.
- Coordinate the other agents and resolve conflicts.
- Measure whether adaptations helped and learn from the result.

Inputs: behavior events, assessment results, plan outcomes, user feedback.
Outputs: updated Personal Model, risk predictions, strategy directives for other agents.

Example adaptation rules

Condition detected	Adaptation
High risk of missing a large task on a heavy-lecture day	Split into smaller sessions in advance and schedule in productive windows
Task type consistently takes longer than estimated	Apply a correction factor to future estimates
Repeated errors in a topic	Request a targeted review plan and extra practice
Declining adherence for several days	Reduce daily load and switch the coach to a supportive tone

9.2 Personal Planning Agent

Role: Turns courses, deadlines, goals, and the Personal Model into concrete daily and weekly plans.

Responsibilities

- Generate daily and weekly schedules from fixed commitments and deadlines.
- Apply estimation corrections and productivity windows from the Personal Model.
- Split large tasks and place buffers before deadlines.

- Re-plan when events change, following strategy directives from the Manager.

Inputs: timetable, exams, assignments, goals, Personal Model, directives.
Outputs: structured plan (tasks, time slots, priorities, rationale).

Design note: each scheduled item includes a short *rationale* explaining why it was placed there, which improves transparency and user trust.

9.3 Assessment Agent

Role: AI tutor and evaluator that generates learning material and measures understanding.

Responsibilities

- Generate quizzes, mock exams, and flashcards for a course and topic.
- Grade answers and explain mistakes.
- Classify errors (concept gap, careless error, memorization, time pressure).
- Identify weak topics and report them to the Manager.
- Adjust difficulty based on the user's current level.

Inputs: course material or topic, difficulty, Personal Model.
Outputs: assessments, scores, error analysis, weak-topic list.

9.4 Goal & Life Coach Agent

Role: Supports academic, career, and personal goals and keeps the user accountable.

Responsibilities

- Help the user define clear, measurable goals and milestones.
- Break long-term goals into weekly actions that the Planning Agent can schedule.
- Provide motivation, reflection prompts, and weekly reviews.
- Ask for reasons when plans are not followed and report them to the Manager.
- Manage streaks, achievements, and accountability messages.

Inputs: goals, progress, adherence data, Personal Model.
Outputs: milestone plans, coaching messages, reflection summaries, gamification updates.

Boundary: the coach offers productivity and study guidance only. It does not provide medical or clinical advice and directs users to professional help if serious distress is expressed.

9.5 Generative LLM Layer

Role: The shared generative engine used by all agents for reasoning and language generation.

Responsibilities

- Generate plans, explanations, quizzes, flashcards, and coaching messages.
- Return **structured JSON** for plans and assessments so the application can validate and store them.
- Receive a compact, privacy-aware summary of the Personal Model as context.

Controls

- Prompt templates per agent with fixed schemas.
- Output validation and retry on malformed responses.
- Rate limiting and cost control.
- API keys kept on the server side only.

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10. Struggle Prediction (Advanced Feature)

The Struggle Predictor estimates, ahead of time, the likelihood that the user will fail to complete a planned task or perform poorly on a topic.

10.1 Predicted Targets

1. **Task-miss risk:** probability that a scheduled task is missed or deferred.
2. **Topic-struggle risk:** probability that the user scores poorly on a topic.
3. **Overload risk:** probability that a day's plan is unrealistic.

10.2 Features

Feature group	Examples
Task features	Task type, estimated duration, priority, days to deadline
Context features	Lectures that day, total planned load, time of day
User features	Estimation bias, deferral rate for this task type, productivity at this hour
Learning features	Recent quiz scores, error rate on the topic, time since last review

10.3 Approach

- **Phase 1 (baseline):** transparent rule-based and weighted scoring using Personal Model features.
- **Phase 2:** lightweight statistical model (logistic regression or gradient boosting) trained on accumulated events, compared against the baseline.
- The risk score is mapped to **Low / Medium / High**, and High risk triggers a preventive adaptation.

10.3.1 Data for Evaluation

Because a new project has little real data, evaluation uses two sources: (a) real usage logs from pilot users, and (b) **simulated user personas** (for example: night-owl procrastinator, steady morning worker, overloaded commuter) that generate realistic behavior traces for controlled experiments.

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11. Assessment and Study Flow

1. The user selects a course and topic, or the Planning Agent schedules a study session.
2. The Assessment Agent generates a quiz, flashcards, or exam at the right difficulty.
3. The user answers; the agent grades and explains.
4. Errors are classified and weak topics are identified.
5. Results are sent to the Adaptive Manager.
6. The Manager updates the Personal Model and the **learning strategy** (for example, more practice questions, shorter review intervals).
7. The Planning Agent adds targeted review sessions to the next plan.

This closes the loop between learning results and planning.

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12. Data Model (Cloud Firestore)

All data lives in Firestore. Profile images are resized and stored as a compressed Base64 string inside the user document, so Firebase Storage is not required.

Collection / Path	Purpose	Key fields
`users/{uid}`	Profile and onboarding data	name, education, college, semester, photo (Base64), createdAt
`users/{uid}/courses/{id}`	Courses per semester	name, semester, credits, schedule
`users/{uid}/deadlines/{id}`	Exams and assignments	type, courseId, dueDate, weight
`users/{uid}/goals/{id}`	Academic, career, personal goals	title, category, target, milestones, status
`users/{uid}/tasks/{id}`	Planned tasks	title, type, estimate, actual, status, scheduledAt, rationale
`users/{uid}/events/{id}`	Behavior events	eventType, taskId, timestamp, metadata
`users/{uid}/model/current`	Personal Adaptive Model	features, confidence, updatedAt
`users/{uid}/predictions/{id}`	Risk predictions and outcomes	target, riskScore, level, outcome
`users/{uid}/assessments/{id}`	Quizzes, exams, flashcards	topic, questions, answers, score, errorTypes
`users/{uid}/gamification/state`	Streaks and achievements	streak, points, badges
`users/{uid}/reports/{id}`	Improvement reports	period, metrics, summary

12.1 Security Rules (Principles)

- A user can read and write only documents under their own `users/{uid}` path.

- Authentication is required for every operation.
- Field validation for types, sizes, and allowed values.
- The LLM API key is never exposed to the client.

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13. User Interface Design

13.1 Design Principles

- Calm, premium dark interface with a deep neutral background and off-white text.
- Electric indigo/violet as the single accent color.
- Lucide line icons; no emoji in the interface.
- Subtle GSAP animation to guide attention without distraction.
- Responsive layout for desktop and mobile browsers.

13.2 Main Screens

Screen	Purpose
Sign in / Sign up	Firebase authentication
Onboarding	Education, college, courses per semester, profile photo, goals
Dashboard	Today's plan, risk alerts, streak, quick actions
Planner	Weekly view with task rationale and manual edits
Study Room	AI tutor, quizzes, flashcards, exams
Goals	Goal tracking, milestones, weekly review
My Model	Transparent view of what the system has learned about the user
Reports	Progress and adaptive model improvement reports

13.3 Transparency and Control

The **My Model** screen shows the learned insights in plain language (for example, "You complete most tasks between 8 and 11 AM") and lets the user correct or reset them. Adaptation is explainable, not a black box.

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14. Evaluation of the Adaptive Model (Option C)

14.1 Evaluation Questions

1. Does the Personal Model become more accurate as data accumulates?
2. Do predictions identify real struggles before they happen?
3. Do adaptations improve outcomes compared with a non-adaptive baseline?

14.2 Metrics

Area	Metric	Meaning
Estimation	Mean Absolute Percentage Error of task-duration estimates	Lower is better as the model learns
Prediction	Precision, Recall, F1, ROC-AUC of struggle prediction	Quality of early warnings
Prediction	Brier score / calibration	Whether risk scores reflect real probabilities
Planning	Plan adherence rate	Completed tasks divided by planned tasks
Planning	Deferral rate	Share of tasks postponed
Learning	Score improvement on weak topics	Effect of adapted learning strategy
Engagement	Streak length, weekly active days	Consistency support
Adaptation	Preventive-action success rate	Share of high-risk tasks completed after adaptation

14.3 Experimental Design

- **A/B comparison:** *Static planner* (no learning) vs. *Adaptive system* on the same user personas or the same pilot users over time.
- **Learning curves:** metrics tracked week by week to show improvement as the model gains data.
- **Ablation tests:** disable one Personal Model dimension at a time to measure its contribution.
- **Baseline vs. Phase 2 predictor:** rule-based versus statistical model.

14.4 Improvement Reports

The system generates periodic reports containing:

- Summary of the user's behavior and strengths.
- Changes in adherence, estimation accuracy, and quiz performance.
- Predictions made, outcomes, and which adaptations helped.
- Recommended next strategy adjustments.

An aggregate evaluation report (all personas and pilot users) is included in the repository for the final presentation.

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15. System Testing

Test type	Scope	Examples
Unit tests	Pure logic	Duration correction, risk scoring, streak calculation, schedule conflict detection
Agent tests	Each agent in isolation	Planning output matches schema and respects deadlines; Assessment returns valid questions with correct answer keys
Integration tests	Agent-to-agent flows	Quiz result updates the Personal Model and produces a review task
LLM output tests	Generative reliability	JSON schema validity rate, retry behavior, hallucination spot checks
Security tests	Firestore rules	A user cannot read or write another user's data
UI tests	Core user journeys	Sign up, onboarding, create plan, complete quiz
Simulation tests	End-to-end adaptivity	Persona runs for several simulated weeks with metrics collected
Usability tests	Pilot users	Task success and satisfaction questionnaire

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16. Non-Functional Requirements

- **Privacy:** only the data needed for coaching is collected; users can view, correct, export, and delete their data.
- **Security:** authenticated access, strict Firestore rules, server-side API keys.
- **Reliability:** graceful fallback to rule-based planning if the LLM is unavailable.

- **Performance:** plan generation should feel responsive; heavy computation runs asynchronously with progress feedback.
- **Explainability:** every plan item and every adaptation includes a short reason.
- **Maintainability:** modular JavaScript files, clear folder structure, documented prompts.

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17. Risks and Mitigations

Risk	Impact	Mitigation
Cold start: little data at first	Weak personalization	Onboarding questions, defaults, confidence scores, fast early learning
LLM hallucination or malformed output	Wrong quiz answers or invalid plans	Structured schemas, validation, answer-key checks, retries
Over-adaptation to noisy data	Unstable plans	Moving averages, minimum evidence thresholds, user override
Privacy concerns	User distrust	Transparency screen, minimal data, deletion and export
API cost and latency	Slow or expensive use	Caching, compact context, rate limits
Limited real data for evaluation	Weak evidence	Simulated personas plus pilot users
Scope too large	Unfinished work	Phased roadmap with a working MVP first

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18. Implementation Roadmap

Phase	Focus	Deliverables
1	Foundation	Authentication, onboarding, Firestore structure and security rules
2	Planning core	Personal Planning Agent, task tracking, behavior event logging

Phase	Focus	Deliverables
3	Learning core	Assessment Agent, quizzes, flashcards, exams, error analysis
4	Adaptivity	Personal Model, Adaptive Manager, strategy rules
5	Prediction	Struggle Predictor (baseline then statistical model)
6	Coaching	Goal & Life Coach, gamification, weekly review
7	Evaluation	Simulation, metrics, A/B and ablation, improvement reports, tests
8	Delivery	Documentation, proposal, presentation, final GitHub release

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19. GitHub Repository Structure

```
adaptive-ai-study-coach/
├── README.md
├── docs/
│   ├── project-documentation.pdf
│   ├── proposal.pdf
│   └── presentation.pptx
├── src/
│   ├── index.html
│   ├── css/
│   ├── js/
│   │   ├── firebase.js
│   │   ├── agents/
│   │   │   ├── adaptiveManager.js
│   │   │   ├── planningAgent.js
│   │   │   ├── assessmentAgent.js
│   │   │   └── coachAgent.js
│   │   ├── model/
│   │   │   ├── personalModel.js
│   │   │   └── predictor.js
│   │   └── ui/
│   └── assets/
├── backend/            # LLM proxy / serverless function
├── evaluation/
│   ├── personas/
```

```
|   ├── simulate.py
|   └── reports/
├── tests/
└── firestore.rules
```

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20. Conclusion

The Adaptive AI Personal Life & Study Coach moves beyond generic planners by learning how each user actually works. Its Personal Adaptive Model, the cooperating agents, and the Collect → Understand → Predict → Adapt → Execute → Measure → Learn loop together create a system that anticipates problems and adjusts its strategy. The Option C extensions — struggle prediction, rigorous evaluation of the adaptive model, system testing, and improvement reports — make the project measurable and verifiable rather than only conceptual.

Appendix A — Glossary

Term	Meaning
Personal Adaptive Model	Per-user model of productivity, focus, deferral, estimation, and learning patterns
Strategy adaptation	Changing how the system plans and teaches, not just when
Struggle prediction	Estimating likelihood of failure before it happens
Cold start	Initial period with little user data
Ablation	Removing one component to measure its contribution
USP	Unique Selling Proposition